A compactified functional pencil and a two-sided Einstein energy window

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This note extends the finite-parameter non-saturation theorem to its vanishing-Weyl endpoint, with an explicit degenerating weight. It then obtains the upper, as well as the lower, quantitative energy envelope. The analytic input is the existing bounded-form compatibility theorem; the classical feasible energy interval is not a new curvature inequality.

1. Uniformity on the unbounded parameter interval

Let (M, g) be closed, connected, oriented and four-dimensional. Fix an actual Spin^c structure whose ordinary Seiberg–Witten equations are solvable for every metric conformal to g. Write

$$p = (a_g^+)^2 > 0, \quad V = \mathrm{Vol}(g), \quad r = \fint |\mathrm{Ric}_+|_g^2, \quad b_a = (-a^2)_+/V,$$

$$D_\beta = \frac{\beta+1}{\beta}\int (s^2 + 6\beta|W^+|^2) - 72\pi^2 p, \qquad \beta \geqslant 2.$$

The conventions are tensor Ricci norm and Hilbert–Schmidt Weyl endomorphism norm. The class is fixed while the conformal weight varies.

Theorem 1. There is a universal $c > 0$ such that, for every finite $\beta \geqslant 2$,

$$\boxed{(1 + b_a)(r + D_\beta/V) \geqslant c\frac{(\beta-2)^2}{\beta^4}.} \tag{1}$$

If $a^2 \geqslant T = 2\chi + 3\tau$, the factor $1 + b_a$ can be removed, after reducing the universal constant. Unlike the earlier coefficient c_L, the constant in (1) does not depend on a parameter cutoff. Its explicit weight vanishes at both the Kähler endpoint and the vanishing-Weyl endpoint.

Proof. Put $w = \sqrt{6}|W^+|$, $q = -s + w$, $e_s = s + 4$, $z = s + \beta w$, and $\kappa = r + D_\beta/V$. The exact nonnegative decomposition in the finite-parameter proof, Section 2, implies

$$\fint e_s^2 \leqslant 4r, \quad \fint z^2 \leqslant \beta D_\beta/V, \quad \Delta_+/V \leqslant D_\beta/V.$$

Consequently

$$\fint (w - 4/\beta)^2 \leqslant 2D_\beta/(\beta V) + 8r/\beta^2 \leqslant 3\kappa,$$

$$\fint (q - q_\beta)^2 \leqslant 14\kappa, \qquad q_\beta = 4 + 4/\beta \in [4, 6]. \tag{2}$$

In particular the normalized Ricci, scalar and W^+ squared energies are bounded by $C(1 + \kappa)$, uniformly for $\beta \geqslant 2$.

Here is the required tracking of the form-production constants; uniformity is not inferred merely by sending L to infinity. The truncated weight $Q = \max(q, 3)$, $f_0 = \max(\sqrt{Q}, (-s)_+/4)$ is independent of β. The inherited moment argument gives the same bounds $|\psi| \leqslant 4$, $D_0 \leqslant 4S/27 + o(1)$ and the same amplitude control, with $S \leqslant 117V\kappa$. Its errors involve only e_s, Δ_+ and Q. For the normalized form and Weyl model the coefficients are

$$c_\beta^2 = \frac{9\beta}{8(\beta+1)} \in [3/4, 9/8], \quad \alpha_\beta = \frac{9}{8(\beta+1)} \leqslant 3/8, \quad t = 1/\beta \in (0, 1/2].$$

Set $\omega = c_\beta\psi$ and $A_{\omega,t} = t(|\omega|^2 I/3 - \omega\otimes\omega)$. The polynomial fit identity is

$$|W^+ - A_{\omega,t}|^2 = \tfrac{1}{6}(w - 2\alpha_\beta|\psi|^2)^2 + 2\alpha_\beta E_\lambda.$$

The amplitude estimate, (2), and $2\alpha_\beta|\psi|^2 = u^2/(\beta+1)$ therefore yield, with one smooth actual solution and an arbitrarily small allowance,

$$\|\omega\|_\infty \leqslant 3\sqrt{2}, \quad d_t := \fint \left(|\operatorname{Ric} + g|^2 + |W^+ - A_{\omega,t}|^2 + (|\omega|^2 - 2)^2\right) \leqslant C\kappa + o(1). \tag{3}$$

No inverse power of t is used here. The determinant energy identity and the bounded-spinor Hessian estimate then give

$$B_\omega := \fint |\Delta_H\omega|^2 \leqslant C(1 + \kappa + b_a) + o(1), \tag{4}$$

again uniformly: the needed curvature energies are bounded by (2), and the normalization of the quadratic spinor map is bounded above. This is exactly the estimate proved in Sections 3–5 of the finite-parameter paper, with every parameter-dependent coefficient now bounded explicitly.

The bounded-form compatibility theorem, with the fixed parameter range $0 \leqslant t \leqslant 1/2$, gives

$$t^2(1-2t)^2 \leqslant C\kappa(1 + \kappa + b_a). \tag{5}$$

For $\kappa \leqslant 1$ this proves (1); for $\kappa > 1$ use $t^2(1-2t)^2 \leqslant 1/64$. Let the smoothing allowance tend to zero. If $a^2 \geqslant T$, Chern–Weil gives $b_a \leqslant r/(8\pi^2)$, and the same split removes the square-cost factor. This proves the theorem. No derivative estimate for the harmonic projection is asserted uniformly as $t \to 0$. □

2. A pencil with an actual endpoint at zero

Define on $0 \leqslant t \leqslant 1/2$

$$P_t = t(1+t)\int s^2 + 6(1+t)\int |W^+|^2 - 72\pi^2 tp. \tag{6}$$

For $t > 0$, this is $tD_{1/t}$; at zero it is exactly $P_0 = 6\int |W^+|^2$. Hence $P_t \geqslant 0$ on the closed interval. Multiplying (1) by t, and using $r + tD/V \geqslant t(r + D/V)$, proves

$$\boxed{(1 + b_a)(r + P_t/V) \geqslant ct^3(1-2t)^2.} \tag{7}$$

The same square-cost removal applies when $a^2 \geqslant T$. At $t = 0$ the inequality is trivial and supplies no nonzero lower amplitude or actual anti-self-dual Einstein existence. At $t = 1/2$, the endpoint includes negative Kähler–Einstein metrics. The cubic decay in (7) is a proved sufficient rate from this transfer, not an optimality claim.

For a normalized negative Kähler–Einstein metric and its canonical class,

$$P_t/V = 4(1-2t)^2.$$

This provides an actual normalization check. It does not establish sharpness at the other endpoint.

3. The missing upper energy envelope

Assume now that $A > 0$,

$$T = 2\chi + 3\tau = A(1-h), \qquad 1/4 < h < 1/3, \tag{8}$$

and for every metric one can choose an actual conformally solvable class with $p \geqslant A$ and $a^2 \geqslant T$. Put

$$e = \frac{\int |\operatorname{Ric}|^2}{8\pi^2 A}, \quad F(h) = \frac{(4-\sqrt{1-3h})^2}{9}, \quad G(h) = \frac{(4+\sqrt{1-3h})^2}{9},$$

$$t_*(h) = \frac{1-2\sqrt{1-3h}}{8+2\sqrt{1-3h}} \in (0, 1/8), \qquad \Theta(h) = t_*^3(1-2t_*)^2. \tag{9}$$

Use the all-sign relative defect

$$\mathscr{E}(g) = \inf_{\lambda \neq 0} \frac{\int |\operatorname{Ric} - \lambda g|^2}{\lambda^2 V}.$$

Theorem 2. There is a universal $c_* > 0$ such that every smooth metric under (8) satisfies

$$\boxed{\mathscr{E}(g) + (G(h) - e)_+ \geqslant c_* \Theta(h).} \tag{10}$$

Combining with the inherited lower-envelope theorem, every Einstein candidate must satisfy the strictly smaller energy window

$$\boxed{F(h) + \varepsilon h^2 \leqslant e \leqslant G(h) - c_* \Theta(h).} \tag{11}$$

Ricci-flat and positive Einstein metrics are independently excluded by the positive actual monopole scalar estimate. Equation (11) concerns all remaining Einstein metrics, not just minimizers or a gluing neighborhood.

Proof. Write $R = \int |\operatorname{Ric}|^2$, $P = \int |\operatorname{Ric}^\circ|^2$, $C_A = 8\pi^2 A$. The exact Chern–Weil reduction of (6) is

$$P_t = k(t)R + \ell(t)P + 12\pi^2(1+t)T - 72\pi^2 tp,$$

$$k(t) = \frac{(1+t)(8t-1)}{2}, \qquad \ell(t) = 2(1+t)(1-2t). \tag{12}$$

For $t = t_*(h)$, direct substitution gives

$$k(t)G(h) + \tfrac{3}{2}(1+t)(1-h) - 9t = 0,$$

and therefore

$$P_t = C_A k(t)(e - G(h)) + \ell(t)P - 72\pi^2 t(p - A) \leqslant C_A |k(t)|(G(h) - e)_+ + \ell(t)P. \tag{13}$$

This is the opposite sign of the coefficient in the lower-envelope argument. It is why the finite-parameter range beyond $\beta = 8$ controls an upper energy wall.

First fix the target to $\operatorname{Ric} = -g$ and $r = \fint |\operatorname{Ric} + g|^2 \leqslant 1$. The scalar monopole bound and the trace estimate imply

$$C_A/V \leqslant (2 + \sqrt{r})^2 \leqslant 9, \qquad P/V \leqslant r.$$

On $0 \leqslant t \leqslant 1/8$, $|k(t)| \leqslant 1/2$ and $0 \leqslant \ell(t) \leqslant 2$. Using (7), with its square-cost factor removed, in (13) gives

$$c\Theta(h) \leqslant 3r + \tfrac{9}{2}(G(h) - e)_+ \leqslant \tfrac{9}{2}\{r + (G(h) - e)_+\}. \tag{14}$$

For $r > 1$ the desired estimate follows by decreasing the constant. Constant rescaling and taking the infimum over negative target constants prove (10) for the negative-target infimum $\mathscr{E}_-$.

The passage to all signs needs a different argument from the earlier lower-envelope transfer: a very large e now makes the energy excess zero. Here the upper Chern–Weil budget closes that possibility. Let $I = \int s$. If $I \leqslant 0$, direct minimization gives $\mathscr{E}_+ = 4$. If $I > 0$, it gives $\mathscr{E}_+ = \mathscr{E} = 4 - I^2/(RV)$. The signed scalar estimate $\int s_\pm^2 \geqslant 4C_A$ implies

$$\mathscr{E}_+ \geqslant 4/e. \tag{15}$$

If $\mathscr{E}_+ < 1/2$, the chirality upper bound gives

$$e(1 - \mathscr{E}_+) \leqslant U = 3\min(2\chi + 3\tau, 2\chi - 3\tau)/A \leqslant 3,$$

so $e \leqslant 6$. Equation (15) would imply $\mathscr{E}_+ \geqslant 2/3$, a contradiction. Thus $\mathscr{E}_+ \geqslant 1/2$. Since Θ is bounded, this proves (10) also for the positive target infimum, after reducing c_*; take the minimum of the two signwise bounds. □

4. Geometry of the two endpoints and an actual application

The classical Einstein constraints are, with nonnegative x, y,

$$x^2 = 4e, \quad x^2 + 8y^2 = 12(1 - h), \quad x \geqslant 2, \quad x + y \geqslant 3.$$

When $1/4 < h < 1/3$, the ellipse has $x < 3$. Solving these inequalities gives the full classical interval $F(h) \leqslant e \leqslant G(h)$. Thus the interval itself is classical; the positive quantitative margins in (11) are the new application of the differential non-saturation mechanism.

At $h \downarrow 1/4$, the upper optimal parameter tends to $t = 0$, and $\Theta(h) \sim (8/27)(h - 1/4)^3$. At $h \uparrow 1/3$, both optimal parameters meet $t = 1/8$; the interval collapses, consistently with the old critical gap. At the lower-envelope endpoint $h \downarrow 0$, the optimal parameter instead tends to the allowed Kähler value $t = 1/2$. These are different geometric degeneracies; they cannot be treated as one compact parameter range with a fixed forbidden distance.

For an actual example, let

$$M = (\Sigma_2 \times \Sigma_3)\#5\overline{\mathbb{CP}}^2.$$

Its polarized actual classes give $A = 16$, $h = 5/16$, $a^2 = T = 11$. Here

$$F = 25/16, \qquad G = 289/144, \qquad t_* = 1/17, \qquad \Theta = 225/17^5.$$

Any Einstein metric, if one exists, must consequently satisfy

$$\frac{25}{16} + \varepsilon\frac{25}{256} \leqslant \frac{R}{128\pi^2} \leqslant \frac{289}{144} - c_*\frac{225}{17^5}. \tag{16}$$

This is not a nonexistence claim for this example. Its bare two-chirality upper bound is only $e \leqslant 33/16$, so the upper envelope contains information not supplied by that bound alone. The actual monopole selection is the same established polarized input as in the main paper; no new gauge existence theorem is asserted here.

5. Relation to the research goal

The first-line theorem now supplies a two-sided restriction on the entire energy spectrum of possible Einstein outputs in the displayed regime. It does not merely study an energy-minimizing sequence. For a pointwise input, $\mathscr{E} \leqslant 4\delta_\sigma^2$ converts (10) into the corresponding upper-energy constraint on that input; a small residual alone does not force the input energy into a forbidden region.

This note neither classifies the allowed spectra of all Einstein metrics nor constructs an Einstein metric. The separate Bauer–Furuta application in this increment is the new topological-category result. The present contribution is the closed-parameter formulation and its upper-wall application, using the existing analytic mechanism.

Sources: LeBrun's scalar–Weyl inequalities and Chern–Weil reduction; the local functional-pencil proof, Sections 2–6; bounded-form compatibility, Section 6; lower energy envelope; and two-chirality upper bound.

`check_line1_compactified_pencil.py` verifies the exact coefficients, both envelope roots, endpoint rates, actual example, and several deliberate sign errors. These checks do not substitute for the written uniform analytic estimates or actual gauge-solvability inputs.